Algorithms Homework #1

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1 Problem 1

1.a

Two for loops are defined. The first for loop iterates from 1 to n, because of this it runs n iterations. For the inner loop, since it also runs from 1 to n *but* it only runs with an interval of +2, it only runs $\frac{n}{2}$ times. The time complexity equation would be:

$$T(n) = n \times \frac{n}{2} \Rightarrow \frac{1}{2}n^2$$

In big-O notation this would be $O(n^2)$.

1.b

Only one for loop is defined iterating from 1 to n but incrementing by a factor of 3. Because it's incrementing by a multiplicative factor rather than addition, it's run times are measured with $3^k = n$ which is equivalent to $k = \log_3 n$. Therefor the time complexity is given by:

$$T(n) = c \times \log_3 n$$

2 Problem 2

2.a

$$T(n) = 3n^2 + 4n + 10, T(n) = O(n^2)$$

$$3n^2 + 4n + 10 \leq 3n^2 + 4n^2 + 10n^2$$

$$\Rightarrow 3n^2 + 4n + 10 \leq 17n^2 \quad \checkmark \text{ is true}$$

$$\Rightarrow -14n^2 + 4n + 10 \leq 0$$

$$n_0 \geq 0$$

Because $3n^2 + 4n + 10$ is less than $17n^2$ it proves that T(n) is $O(n^2)$ for all values where $n \geq 0$.

2.b

$$T(n) = 3n^2 + 4n + 10, T(n) = \Omega(n^2)$$

$$\Rightarrow 3n^2 + 4n + 10 \geq 3n^2 + 4n^2 + 10n^2$$

$$\Rightarrow 3n^2 + 4n + 10 \geq 17n^2 \quad ✗ \text{ is not true}$$

Because $3n^2 + 4n + 10$ is not greater than $17n^2$, T(n) can not have a best case scenario of $\Omega(n^2)$ at any value of $n_0 \geq 0$.

2.c

$$T(n) = 3n^2 + 4n + 10, T(n) = \Theta(n^2)$$

$$\Rightarrow 3n^2 + 4n^2 + 10n^2 \leq 3n^2 + 4n + 10 \leq 3n^2 + 4n^2 + 10n^2$$

$$\Rightarrow 3n^2 + 4n + 10 \neq 17n^2 \quad ✗ \text{ is not true becase } 3n^2 + 4n + 10 \geq 17n^2 \text{ is not true}$$

Because $3n^2 + 4n_{10}$ is not greater than $17n^2$ even though it is lesser than it, it isn't true that $3n^2 + 4n + 10$ has an exact order of $\Theta(n^2)$.

2.d

$$T(n) = 3n \log n - 2n + 7, T(n) = \Omega(n \log n) \text{ assuming base of 2}$$

$$\Rightarrow 3n \log n - 2n + 7 \geq 3n \log n - 2n \log + 7n \log n$$

$$\Rightarrow 3n \log n - 2n + 7 \geq 8n \log n \quad ✗ \text{ not true}$$

$$\Rightarrow n_0 = 0 ✓ \text{ but } n_0 > 0 ✗ \text{ not true}$$

Because $3n \log n - 2n + 7$ is not greater than $8n \log n$ at any value greater than zero, it can not be said that $T(n) = \Omega(n \log n)$.

3 Problem 3

3.a

$$T(n) = T\left(\frac{n}{2}\right) + 5$$

$$\Rightarrow f(n) = 5, a = 1, b = 2$$

$$\Rightarrow 5 = n^{\log_2 1} \Rightarrow 5 = n^0 = 1 \text{ therefor both are constant}$$

$$T(n) = \Theta\left(n^{\log_2 1} \log n\right) \Rightarrow T(n) = \Theta(n^0 \log n) \Rightarrow T(n) = \Theta(\log n)$$

3.b

$$T(n) = 2T\left(\frac{n}{2}\right) + n^2$$

$$\Rightarrow a = 2, b = 2, f(n) = n^2$$

$$\Rightarrow n^{\log_2 2} = n^1 = n, n \neq n^2, n < n^2$$

$$\text{Because } \; n^{\log_b a} < f(n), T(n) = \Theta(f(n)) \Rightarrow T(n) = \Theta(n^2)$$

3.c

$$T(n) = T\left(2\frac{n}{3}\right) + n$$

$$a = 1, b = \frac{3}{2}, f(n) = n$$

$$n^{\log_{\frac{3}{2}} 1} \Rightarrow n^0 \Rightarrow 1, f(n) = n > 1$$

$$\text{Because } \; f(n) > n^{\log_b a}, T(n) = \Theta(f(n)) \Rightarrow T(n) = \Theta(n)$$

3.d

$$T(n) = 8T\left(\frac{n}{2}\right) + n^3$$

$$a = 8, b = 2, f(n) = n^3$$

$$n^{\log_2 8} = n^3, f(n) = n^3 = n^3$$

$$\text{Because } \; f(n) = n^{\log_b a}, T(n) = \Theta\big(n^{\log_b a} \log(n)\big) \Rightarrow T(n) = \Theta\big(n^3 \log n\big)$$

3.e

$$T(n) = 5T\left(\frac{n}{5}\right) + \sqrt{n}$$

$$a = 5, b = 5, f(n) = \sqrt{n}$$

$$n^{\log_5 5} = n^1 = n, f(n) = \sqrt{n} < n, n_0 > 1$$

$$\text{Because } \; f(n) < n^{\log_b a}, T(n) = \Theta\big(n^{\log_b 1}\big) \Rightarrow T(n) = n$$

4 Problem 3

4.a

$$a = 4, b = 3, f(n) = O\big(n^2\big) = n^2$$

$$\text{The recurrence equation: } \; T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

$$\Rightarrow T(n) = 4T\left(\frac{n}{3}\right) + n^2$$

4.b

$$n^{\log_3 4} < n^2$$

$$\text{Because } \; n^{\log_b a} < f(n), T(n) = \Theta(f(n)), T(n) = \Theta\big(n^2\big)$$

$$T(k) \leq ck^2 < n$$

$$\text{test}$$